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UNEVEN ACCESS TO QUALITY EDUCATION IN THE DIGITAL

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ABSTRACT

Despite the widespread adoption of digital learning platforms, access to quality online education remains uneven due to constraints such as unreliable internet connectivity, limited device availability, and rigid platform design. Many existing systems fail to adapt to individual learner backgrounds, study schedules, and resource limitations, resulting in reduced engagement and inconsistent learning outcomes. This project presents PathWise, an online learning platform designed with a focus on accessibility, personalization, and practical usability. The system generates customized learning roadmaps based on user-defined goals, prior knowledge, available study time, and learning pace. Learning content is organized dynamically and delivered through lightweight interfaces to support use under low-bandwidth conditions. The platform also integrates structured progress tracking and periodic evaluations to help learners maintain consistency and measure improvement. By prioritizing adaptive system design and real-world constraints, PathWise demonstrates how personalized learning platforms can be engineered to support inclusive and flexible digital education environments.

Keywords: Personalized Learning, Online Education, Accessibility, Learning Roadmap, System Design.

I. INTRODUCTION

The rapid growth of online learning platforms has changed how educational content is accessed and consumed. Learners can now study across geographical boundaries and flexible schedules. However, this expansion has also revealed practical limitations in many existing systems, particularly for users who operate under constrained conditions such as limited internet bandwidth, shared devices, or irregular study time. As a result, the benefits of digital education are not experienced uniformly by all learners.

Most current online learning platforms follow fixed course structures and assume continuous connectivity and a high level of digital familiarity. Such assumptions often overlook the needs of learners who require adaptable pacing, simplified interfaces, or content organization based on their prior knowledge. When these needs are not addressed, learners struggle with consistency, engagement, and long-term progress, even when quality learning resources are available.

From an engineering standpoint, these challenges indicate a gap between system design and real-world usage patterns. Rather than increasing feature complexity, there is a need for platforms that prioritize adaptability, clarity, and efficient resource utilization. Effective personalization should not only recommend content but also structure learning paths in a way that aligns with individual goals, time availability, and learning capacity.

This paper presents PathWise, an online learning platform designed to address these issues through personalized roadmap generation and accessibility-focused design. The system collects user-specific inputs such as learning objectives, existing knowledge level, and daily study duration to generate a structured learning path. Content delivery and progress tracking are designed to function reliably under varying network and device conditions.

The primary contribution of this work lies in demonstrating how thoughtful system design and modular architecture can improve the usability and inclusiveness of digital learning platforms. By grounding design decisions in practical constraints and learner behavior, Path Wise aims to support a wider range of

users and promote more consistent learning outcomes in digital education environments.

II. METHODOLOGY

The methodology adopted for this work focuses on the practical design and implementation of an accessible online learning platform rather than theoretical analysis. The development process begins by identifying real- world constraints faced by learners, including inconsistent internet connectivity, limited access to personal devices, and variations in daily study availability. These constraints directly influence the system design decisions and feature prioritization.

Based on this analysis, the platform requirements are defined in two categories. Functional requirements include user registration and authentication, personalized learning roadmap generation, access to curated learning resources, progress tracking, and assessment management. Non-functional requirements emphasize ease of use, responsiveness under low-bandwidth conditions, system reliability, and scalability to support multiple users without performance degradation.

A modular design approach is adopted to simplify development and future expansion. Core system components such as user management, roadmap generation, content delivery, and analytics are implemented as independent modules. This structure allows individual components to be modified or enhanced without affecting the overall system workflow.

To support learners operating under varying network conditions, the platform organizes learning content dynamically and avoids heavy resource usage. Interface elements are kept minimal to reduce loading time and improve usability across devices. User interactions are designed to be straightforward, ensuring that learners with different levels of technical familiarity can navigate the platform without difficulty.

Overall, this methodology emphasizes practical system development guided by user behavior and environmental constraints. By aligning technical implementation with real usage scenarios, the proposed approach supports the creation of a flexible and inclusive learning platform.

Architecture

The system architecture of PathWise is designed using a modular approach to ensure simplicity, scalability, and ease of maintenance. The platform is structured into distinct layers, each responsible for a specific set of functions, allowing independent development and future enhancements without affecting the overall system.

Users interact with the platform through web and mobile interfaces that serve as the presentation layer. These interfaces are designed to be lightweight and responsive so that the system remains usable across different devices and network conditions. The user interface enables registration, login, roadmap viewing, lesson access, and progress monitoring.

The application logic layer manages core platform operations. This layer handles user input processing, learning roadmap generation, lesson organization, and evaluation workflows. Personalization logic uses user- provided details such as learning goals, existing knowledge level, and available study time to generate and update individualized learning paths. This separation of logic ensures that personalization remains consistent regardless of the device used by the learner.

Data storage is managed through dedicated databases that maintain user profiles, learning progress, roadmap structures, and assessment results. Storing progress data separately allows the system to track learning continuity and support roadmap regeneration when user preferences change.

An analytics component monitors user interactions and learning progress to provide insights into platform usage and learner engagement. These insights support performance evaluation and help identify areas for future system improvement.

The entire platform is deployed on cloud infrastructure to support scalability and availability. Cloud deployment allows the system to handle multiple users efficiently while maintaining reliable access and data security. Overall, the architecture balances modularity, performance, and accessibility to support a personalized learning experience across diverse user environments

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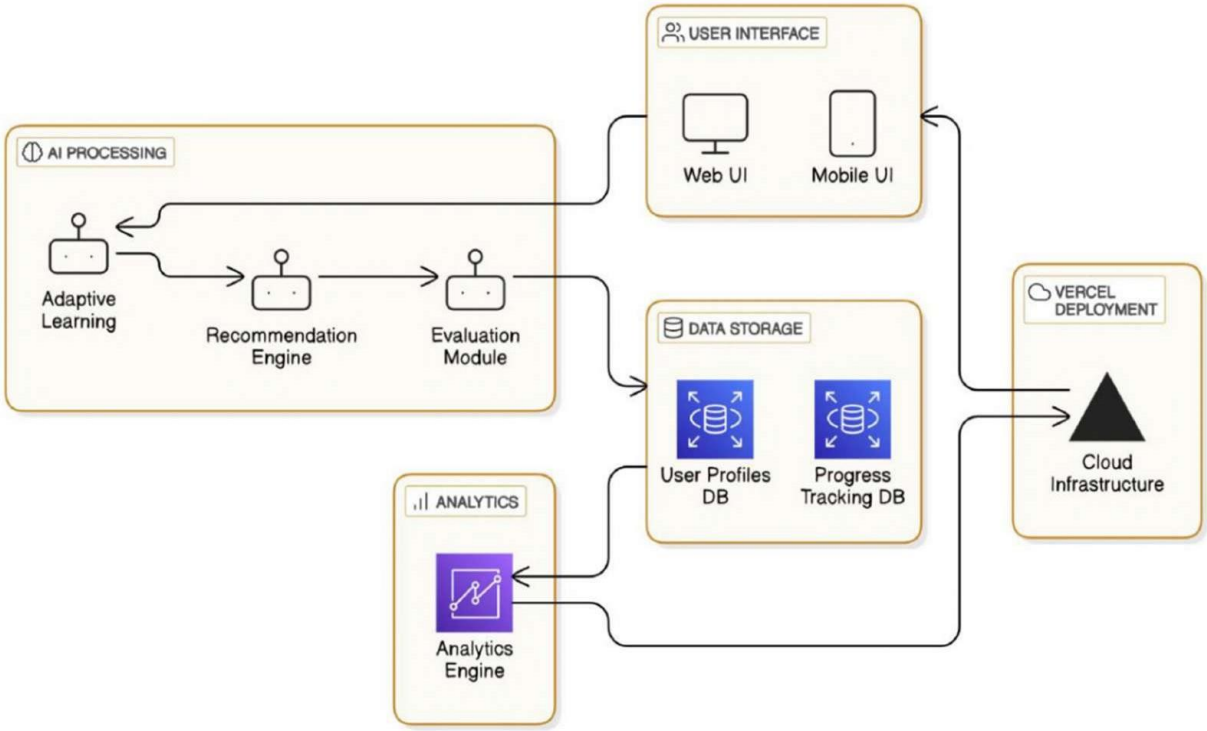

Figure 1: Architecture Diagram

Flow of Project (PathWise)

The workflow of the PathWise platform begins with user onboarding. Learners first register and authenticate themselves to gain access to the system. Once logged in, they are directed to a personalized dashboard that acts as the primary interaction point for all learning activities.

From the dashboard, users complete a structured question-and-answer interaction designed to capture essential learning details such as educational goals, current knowledge level, preferred learning pace, and available daily study time. This information forms the basis for all subsequent personalization within the platform.

Using the collected inputs, the system generates a customized learning roadmap that defines a clear sequence of topics and learning milestones. The roadmap is designed to provide direction and structure while remaining flexible to accommodate individual learning needs. Relevant learning materials are then organized and mapped to each stage of the roadmap.

Learning content is delivered in a structured manner, supported by concise explanatory notes and supplementary resources. Periodic assessments in the form of quizzes and assignments are integrated into the learning flow to evaluate understanding and reinforce concepts. Lesson completion and assessment results are recorded to track user progress.

Progress is continuously monitored, and users can update their goals or preferences at any time. When changes are made, the system regenerates the learning roadmap and adjusts content sequencing accordingly. This adaptive flow ensures that the learning experience remains aligned with user progress and evolving objectives.

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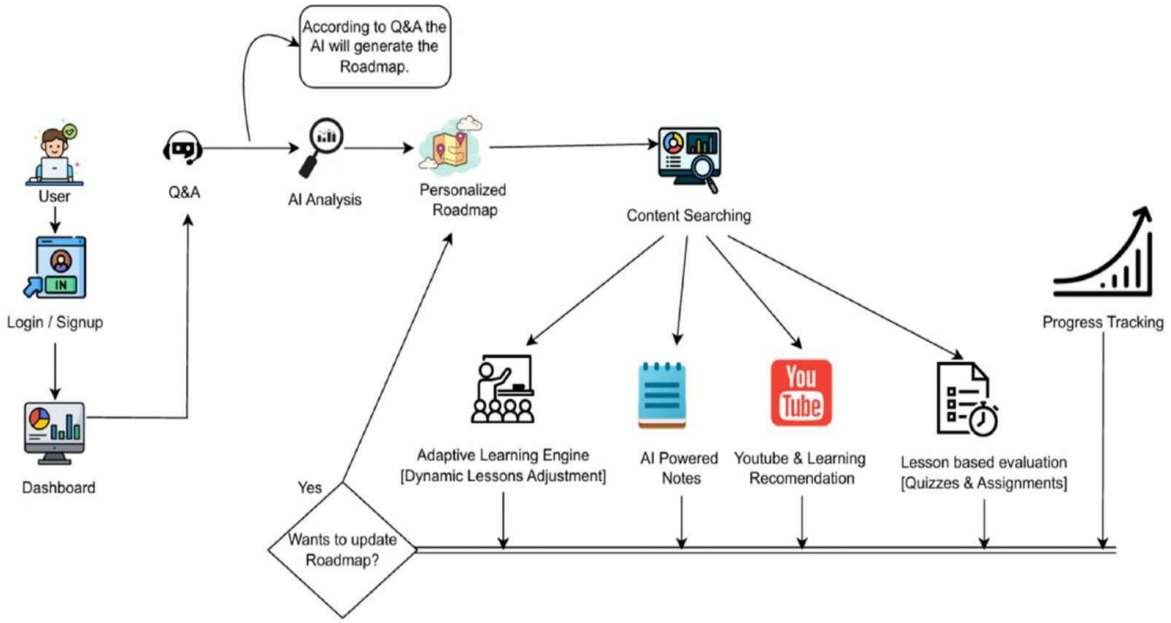

Figure 2: Flow Diagram

III. RESULTS

Table 1. Sample Test Cases for PathWise Platform

ID	Test Description	Expected Result	Actual Result	Status
TC01	Verify Supabase authentication with valid credentials	Successful login and session token creation	As expected	Pass
TC02	Test AI-based roadmap generation and Supabase sync	Generated roadmap stored accurately in database	As expected	Pass
TC03	Check user profile edits and theme update	Data updated in Supabase and reflected in UI	As expected	Pass
TC04	Validate lesson completion tracking in Supabase	Lessons marked completed and dashboard updated	As expected	Pass
TC05	Evaluate API error handling with invalid input	Appropriate TypeScript validation error displayed	As expected	Pass

IV. CONCLUSION

This work presented PathWise, an online learning platform designed to support personalized and accessible digital learning through practical system design. The platform uses user-provided inputs such as learning goals, existing knowledge level, available study time, and preferred pace to generate structured learning roadmaps that guide learners through a clear and manageable learning path.

The system emphasizes simplicity and adaptability by organizing learning content dynamically and tracking progress at the lesson level. By allowing users to update their goals and preferences, the platform ensures that learning paths remain aligned with individual progress and changing requirements. The modular architecture supports scalability and enables the system to be extended to different learning domains without major structural changes.

Overall, PathWise demonstrates how focused engineering decisions and user-centered design can improve the usability and effectiveness of online learning platforms. By addressing practical constraints

such as varied learning speeds and resource limitations, the system provides a flexible foundation for inclusive digital education.

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